

Case Study-I

Data Literacy in Beşiktaş Square, İstanbul/Türkiye

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CASE STUDY AREA

Beşiktaş Square
Beşiktaş Square is a major urban node where ferry terminals, bus stops, and metro access intersect with recreational spaces, informal gathering areas, and public seating. This overlap creates conflicts between fast-paced pedestrian flows and stationary or slow-moving activities.

PROBLEM DEFINITION

Urban Permeability and Pedestrian Flow Conflict
The intersection of transit-oriented pedestrian routes and recreational zones within Beşiktaş Square leads to spatial congestion, disrupted movement patterns, and underutilized areas. The aim of this study is to analyze pedestrian movement patterns through data-driven methods and to propose a new ground and urban furniture configuration that optimizes pedestrian flow while supporting social and recreational use.

Data Engineer

Design the system that transforms raw urban data.. how urban data is stored, organized, and managed so that it can be reliably used in architectural design.

In Beşiktaş Square, **pedestrian movement is complex** because **ferry, bus, and metro** routes **intersect with public** and recreational areas. To understand this situation, we use **drone footage** as the **main data source**, since it **captures** real pedestrian behavior at an **urban scale**.

Drone footage is **unstructured and high-volume**, so it needs a **clear processing workflow**. The **raw videos** are first collected and **pre-processed** through **stabilization** and **spatial alignment**. Pedestrian movements are then automatically **detected and tracked using ML-based methods**.

Since the study works with **different types of data**, they are stored in different systems. **Structured data**, such as pedestrian counts and time-based data, is stored in **databases or a data warehouse**. **Unstructured data**, such as raw drone footage, is stored in a **data lake**. -- working with multiple data types which means they cannot be stored in a single format or system.

-- To **connect these different data types**, we propose a **data lakehouse approach**, where structured outputs—such as pedestrian density maps and flow lines—are **derived** from unstructured drone data and made accessible for analysis and design.

-- Through this data management structure, **raw visual data is transformed into curated datasets** that architects can use to inform decisions on ground design, urban furniture placement, and spatial organization, while maintaining control over data quality and accessibility.

Data Analyst / Scientist

#UrbanAnalytics #TrajectoryData
#SpatialMetadata #OutlierRemoval
#BehavioralMapping

The Data Dictionary & Feature Extraction

-**Raw Input:** Unstructured UAV video footage.

-**AI Pipeline:** Using Object Detection (e.g., YOLO) for pedestrian bounding boxes and Multi-Object Tracking (e.g., DeepSORT) to extract motion. (Optional: *Semantic Segmentation like SAM to map walkable boundaries*).

-**The Data Dictionary:** We don't save video; we extract a matrix. Coordinates (u, v), scale (r, h), and velocity vectors.

-**Metadata is Crucial:** Trajectory data means nothing without context. Timestamp, weather, and UAV flight altitude.

Data Quality & "Cleaning the Noise"

-**False Positives:** Object detection might read a manhole cover or a shadow as a person. *Fix:* Statistical filtering (removing outlier)

-**Identity Switches:** Tracking algorithms sometimes misassign IDs. *Fix:* Filtering out illogical jumps.

-**Hardware/Environment limits:** UAV vibration from wind or flying too high reduces pixel clarity.

Visualization & Architectural Translation

-**Trajectory Maps**

-**Distribution Heat Maps:** Assessing the square to locate high-density bottlenecks.

-**Walking Speed Heat Maps:** Highlighting fast transit corridors versus slow zones.

-**Final Output for Design:** Exporting 2D density maps into visualization software to optimize urban furniture placement.

Ethics & Compliance

#KVKK #GDPR #DataMinimization
#PrivacyByDesign #SecondaryUse #ControlCreep
#Anonymization #AggregatedData

Risky Data

-Photos/videos with visible faces
-Social media usernames and personal identifiers
-Raw visual data that can be reused later

Key Ethical & Legal Risks

-Over-collection of data (vs. data minimization)
-Secondary use of data for other purposes
-Control creep (flow analysis → surveillance)
-Misuse of personal data

Protection Measures (KVKK / GDPR)

-Collect only necessary data for a clear purpose
-Prefer aggregated outputs (counts, density maps, heatmaps)
-Anonymize sensitive information (blur faces, remove usernames)
-Limit access to raw data

-Keep risky raw files only for a short time

Main Principle

-**Start the analysis with ethics: Privacy by Design**



Image: AI-generated visualization